Experiment 6: Time Series Analysis of Website Traffic

Objective: To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Dataset: A dataset containing daily website traffic data with columns such as `Date`, `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.

Methodology:

1. **Load and Clean Data:**
    - Import the dataset and parse the `Date` column as a datetime object.
    - Handle missing values by interpolation.
2. **Time Series Decomposition:**
    - Decompose the time series data of `Page_Views` to identify trend, seasonality, and residuals.
3. **Visualization:**
    - Plot the time series data for `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.
    - Use moving averages to smooth the time series and highlight trends.
    - Create seasonal plots to visualize patterns over time.
4. **Anomaly Detection:**
    - Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

Sample code

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose



# Load dataset

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')



# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()
```

```
plt.show()


# Moving Average

data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')

plt.show()


# Anomaly Detection

data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

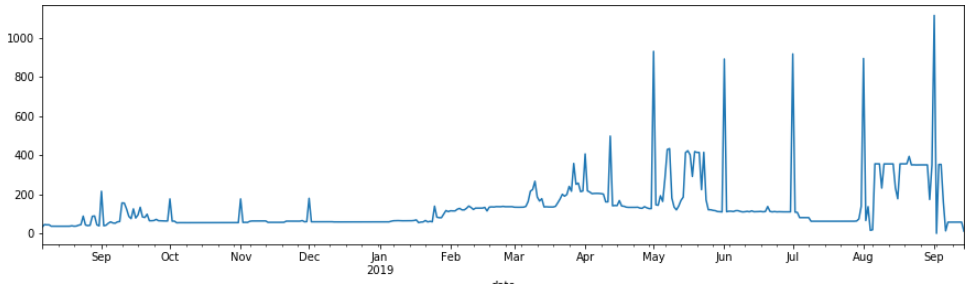
plt.plot(data.index, data['Page_Views'], label='Page Views')

plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')

plt.legend()

plt.show()
```

sample output



Expected Outcomes:

- Clear visualization of trends, seasonality, and anomalies in website traffic.
- Understanding of the underlying patterns in user behavior over time.